mechanical taffy-pulling machine.[25] Those were the golden years of theoretical black-hole research, the 1960s and 1970s! But because the physicists of those golden years were not clever enough at solving Einstein's general relativity equations, one key feature of black-hole behavior eluded them. They could only conjecture that whenever an imploding star creates a singularity, it must also create a surrounding horizon that hides the singularity from view; a singularity can never be created "naked," for all the Universe to see. Penrose called this the "conjecture of cosmic censorship," since, if correct, it would censor all experimental information about singularities: One could never do experiments to test one's theoretical understanding of singularities, unless one were willing to pay the price of entering a black hole, dying while making the measurements, and not even being able to transmit the results back out of the hole as a memorial to one's efforts.

Although Dame Abygaile Lyman, in 2023, finally resolved the issue of whether cosmic censorship is true or not, the resolution is irrelevant to you now. The only singularities charted in your ship's atlases are those inside black holes, and you refuse to pay the price of death to explore them.

Fortunately, outside but near a black-hole horizon there are many phenomena to explore. You are determined to experience those phenomena firsthand and report back to the World Geographic Society, but you cannot experience them near Hades' horizon. The tidal force there is too great. You must explore, instead, a black hole with weaker tidal forces.

General relativity predicts, DAWN reminds you, that as a hole grows more massive, the tidal forces at and above its horizon grow weaker. This seemingly paradoxical behavior has a simple origin: The tidal force is proportional to the hole's mass divided by the cube of its circumference; so as the mass grows, and the horizon circumference grows proportionally, the near-horizon tidal force actually decreases. For a hole weighing a million solar masses, that is, 100,000 times more massive than Hades, the horizon will be 100,000 times larger, and the tidal force there will be 10 billion (10^{10}) times weaker. That would be comfortable; no pain at all! So you begin making plans for the next leg of your voyage: a journey to the nearest million-solar-mass hole listed in Schechter's Black-Hole Atlas—a hole called Sagittario at the center of our Milky Way galaxy, 30,100 light-years away.